

Project 3 — Protein Sequence Similarity Analysis Using BLASTP and COBALT

Overview

This project analyses a 129-residue hen egg-white lysozyme sequence using NCBI Protein BLAST (BLASTP) and COBALT. The goal was to identify similar known protein sequences and examine sequence conservation among related proteins.

Biological Question

Which known protein sequences are most similar to the lysozyme sequence being studied?

Tools Used

NCBI Protein BLAST (BLASTP), NCBI COBALT and protein sequence databases.

Query Sequence

The query was the 129-residue hen egg-white lysozyme sequence used in the earlier structural projects. It begins with KVFGRCELAA.

Method

The sequence was submitted to BLASTP; ranked matches were examined; identity, positives, gaps, E-value and bit score were compared; closely related avian lysozymes were identified; COBALT was opened as a multiple sequence alignment step.

Main BLASTP Results

Hit	Identity	Positives	Gaps	E-value	Score
1LSG_A — HEN EGG WHITE LYSOZYME	129/129 (100%)	129/129 (100%)	0/129	4e-89	266 bits
XP_015711651.2 — lysozyme C	123/129 (95%)	126/129 (97%)	0/129	2e-84	254 bits
ACL81751.1 — lysozyme	123/129 (95%)	124/129 (96%)	0/129	2e-84	254 bits

Interpretation of Top Hit

The strongest visible BLASTP result was 1LSG_A — HEN EGG WHITE LYSOZYME [Gallus gallus]. It showed 129/129 identical residues, 100% identity, 100% positives, no gaps and an E-value of 4e-89. This strongly supports the identity of the query as hen egg-white lysozyme.

Related Avian Sequences

Strong matches also included lysozyme sequences from Coturnix japonica and Bambusicola thoracicus, each showing approximately 95% identity with the query.

Understanding BLAST Results

Identity means exactly matching amino acids. Positives include identical and chemically similar substitutions. Gaps represent inserted or deleted positions in an alignment. E-value estimates the number of similar matches expected by chance; smaller values indicate stronger significance. Bit score summarizes alignment strength; higher values generally indicate stronger matches.

COBALT

COBALT was opened as a multiple sequence alignment step. The returned sequence set contained the query and related lysozyme sequences, but the recorded output displayed “Graphical Overview: Not available”.

Therefore, this project does not claim a detailed COBALT alignment interpretation.

Biological Interpretation

The BLASTP results strongly support the identity of the query as hen egg-white lysozyme. The exact 129-residue match and highly similar avian sequences demonstrate strong sequence conservation.

What I Learned

I learned to interpret sequence identity, positives, gaps, E-values, bit scores and ranked matches, and to distinguish between a computational step that was performed and an analysis whose output was not sufficiently available.

Limitations and Next Step

This project did not include a completed detailed MSA, phylogenetic tree, domain analysis or mapping of conservation onto structure. Future work will examine conserved and variable positions and evolutionary relationships.

Sources

NCBI BLAST • NCBI COBALT • RCSB Protein Data Bank